

## Project Description: Automated API Liveness Testing Script

This project aims to build a **Python-based automated testing script** that verifies the **liveness and availability** of key endpoints from the <https://dummyjson.com> API. The goal is to provide testers with a simple and reliable way to check whether the API endpoints are responsive and returning expected status codes at any time.

The script will cover the following **API categories and endpoints**:

### Products

- **List products (paginated):** <https://dummyjson.com/products>
- **Get single product by ID:** <https://dummyjson.com/products/1>
- **Search products by query:** <https://dummyjson.com/products/search?q=phone>
- **Get products in a category:** <https://dummyjson.com/products/category/smartphones>

### Users

- **List users:** <https://dummyjson.com/users>
- **Get user by ID:** <https://dummyjson.com/users/1>

### Carts

- **List carts:** <https://dummyjson.com/carts>
- **Get cart by ID:** <https://dummyjson.com/carts/1>

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## Goal:

### 1. Automated Testing of Endpoints

- Sends requests to each API endpoint.
- Validates the **HTTP response code** (expects **200 OK**).
- [Advanced-Complex Data Structure Tactic] verifies that the response body contains expected JSON keys (e.g., **products**, **users**, **carts**).

### 2. Reusable & Extensible Script

- Easily add new endpoints to the testing list.
- Centralized configuration for base URL and endpoints.

### 3. User-Friendly Execution

- Tester can run the script with a single command:

```
python test_api_liveness.py
```

- Script outputs a clear summary of results (e.g.,  Pass /  Fail).

#### 4. Optional Enhancements

- Log results to a file (with timestamps).
  - Provide colored console output for readability.
  - Support command-line arguments (e.g., test specific category only).
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#### Example Output (Console)

```
[TESTING API ENDPOINTS]

Products List ..... ✓ PASS (200)
Single Product (id=1) ..... ✓ PASS (200)
Search Products (q=phone) ..... ✓ PASS (200)
Products in Category ..... ✓ PASS (200)

Users List ..... ✓ PASS (200)
User by ID (id=1) ..... ✓ PASS (200)

Carts List ..... ✓ PASS (200)
Cart by ID (id=1) ..... ✓ PASS (200)

All endpoints tested successfully!
```